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I. Introduction

IoT networks now connect household appliances, industrial controllers, and public-service devices that generate markedly different traffic patterns. A compromised endpoint can provide a route to other devices or services. Identifying malicious traffic is therefore only part of the problem. A useful network intrusion detection system (NIDS) must distinguish attack categories, not merely separate benign from malicious flows. CIC-ToN-IoT provides labeled IoT network flows for this multiclass setting, although its heterogeneous features and sharply unequal class frequencies remain challenging for machine-learning models [1], [2].

High dimensionality and class imbalance create related modeling problems. Some traffic features are redundant or carry little signal, adding complexity without improving class separation. Meanwhile, majority classes can dominate training; high accuracy may coexist with poor detection of rare attacks. Accuracy alone hides this. Preprocessing introduces a separate risk. If normalization or other data-dependent steps are applied before splitting, information from held-out flows can enter training and make test results look stronger than they are [3], [4]. An Autoencoder can learn a compact, nonlinear representation [5], [6], while Light Gradient Boosting Machine (LightGBM) is well suited to tabular intrusion data [7]–[9].

Our pipeline begins with all 5,351,760 raw CIC-ToN-IoT flow records. After invalid rows and non-informative fields are removed, 69 traffic features remain. We use a stratified 80:20 train–test split and fit MinMax scaling only on the training partition. The Autoencoder maps the 69 features to a 16-dimensional latent representation, which LightGBM classifies under four training-data treatments: no imbalance handling, class weighting, random upsampling, and random downsampling. Every scenario uses the original, untouched test distribution. We also train LightGBM directly on the 69 normalized features to measure how much discriminative information the latent representation retains.

We make three contributions. First, we implement an end-to-end multiclass NIDS pipeline on the full dataset with explicit controls against preprocessing and test-set leakage. Second, we compare four imbalance-handling scenarios using accuracy, macro precision, macro recall, macro F1-score, per-class metrics, and confusion matrices, which exposes their effects on minority attacks. Third, we contrast the 16-feature latent representation with the original 69-feature representation, quantifying the trade-off between dimensionality reduction and discriminative performance rather than assuming that compression necessarily improves classification.

II. Related Work

Intrusion detection can be organized by what is monitored and how attacks are recognized. Host-based systems inspect activity on individual machines, whereas NIDS analyzes traffic flowing between devices. Signature-based detectors match known patterns and work well when attack definitions already exist; anomaly-based detectors instead flag deviations from learned normal behavior and can expose previously unseen attacks, usually at the cost of more false alarms [10], [11]. For IoT traffic, centralized

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network monitoring avoids requiring an agent on every constrained device [12]. Research has also shifted from binary benign–attack decisions toward multiclass labels that identify attack families [1], [2]. The added detail is operationally useful, but similar traffic signatures and rare classes make the classification problem harder.

Feature reduction addresses the redundancy found in many network attributes. Feature selection retains a subset of the original attributes, preserving their meaning and making model behavior easier to trace [13], [14]. Feature extraction instead transforms the inputs into a new representation [6], [13]. Principal Component Analysis provides a linear projection, whereas Autoencoders can capture nonlinear dependencies; sparse and stacked sparse variants further constrain or deepen that representation [5], [15], [16]. Several intrusion-detection pipelines therefore place an encoder before a separate classifier [17], [18]. We follow that arrangement but retain an original-feature baseline, since low reconstruction error does not guarantee that the latent space preserves every class boundary.

Boosted tree models remain strong candidates for tabular network data. LightGBM builds an ensemble of decision trees efficiently, and prior NIDS studies have combined it with oversampling, feature selection, or optimization [7]–[9], [19]. These combinations matter because imbalance treatments change what the classifier learns: class weights alter each class's contribution without changing the sample set, upsampling adds minority observations, and downsampling removes majority observations [2], [9]. Each choice can trade majority-class performance for minority-class recall. Comparisons on an unchanged test distribution and with macro-averaged metrics are therefore more informative than accuracy alone. Our experiments hold preprocessing, representation, classifier settings, and test data fixed so that differences among the four imbalance scenarios can be attributed to the training-data treatment.

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks …". Instead, try "R. B. G. thanks…". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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